Quantification: 548_hyp_HP_z14c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	25
interior lacuna count	17
frame-edge lacuna count	8

Interior lacunae only: mean, median and sample SD (n = 17)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2772.76	1527	2587.31	px^2
major_axis_length_px	lacuna shape	80.8356	71.3216	39.6731	px
minor_axis_length_px	lacuna shape	42.8181	37.4316	20.5044	px
aspect_ratio	lacuna shape	1.9307	2.0218	0.3976	unitless
eccentricity	lacuna shape	0.8247	0.8691	0.1026	unitless
solidity	lacuna shape	0.7801	0.7951	0.0753	unitless
roots_count	headline	9.6471	7	5.989	count
ring_length_r30_px	headline	308.941	274	187.441	px
ring_length_r60_px	reported	612.177	566	354.98	px
owned_length_px	ownership-dependent	972.654	736.566	846.755	px
edge_count	ownership-dependent	51.1765	27	46.0709	count
mean_edge_length_px	ownership-dependent	19.7153	18.6439	4.5911	px
perimeter_px	normalised	284.341	221.38	176.468	px
ring_area_r30_px2	normalised	8697.24	7467	4434.9	px^2
ring_area_r60_px2	normalised	18584.6	16799	8821.44	px^2
in_frame_fraction_r30	normalised	0.9698	1	0.0664	unitless
in_frame_fraction_r60	normalised	0.9444	1	0.1095	unitless
ring_density_r30	normalised	0.034893	0.0342391	0.007745	px^-1
ring_density_r60	normalised	0.0320149	0.0323261	0.00578502	px^-1
roots_per_100px_perimeter	normalised	3.6451	3.1972	1.5401	per 100 px
ring_attached_length_r30_px	network v2	278.235	243	181.936	px
ring_attached_length_r60_px	network v2	477.118	329	335.808	px
ring_length_w_r30_px	network v2	353.305	318.907	213.775	px
ring_length_w_r60_px	network v2	699.228	643.529	406.652	px
sholl_crossings_r10	network v2	7.0588	5	4.1754	count
sholl_crossings_r20	network v2	8	7	5.0621	count
sholl_crossings_r30	network v2	8.1176	7	5.3954	count
loop_count	loops, report only	3.1765	2	3.2832	count
loop_edge_share	loops, report only	0.1838	0.2347	0.1238	share
ring_width_mean_r30_px	width	7.6601	7.5689	0.5948	px

Per field
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measure	value	units
analysed_area_px2	980048	px^2
total_skeleton_length_px	27553	px
canalicular_length_density_per_px	0.0281139	px^-1
junction_count	597	count
junction_density_per_px2	0.00060915	px^-2
lacuna_count	25	count

lacunae_per_px2	2.551e-05	px^-2
skeleton_component_count	380	count
mean_component_length_px	72.5079	px
median_component_length_px	19	px
field_length_density_w_per_px	0.0320044	px^-1
field_density_without_flagged_per_px	0.0281139	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	248	px
unattached_component_count	14	count
canalicular_length_density_attached_per_px	0.0281139	px^-1
field_length_density_with_unattached_per_px	0.028367	px^-1
loop_count	74	count
loop_edge_share	0.1906	share
edge_count	1548	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8536	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	27553	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2147	91.4001	39.1166	2.3366	0.9038	0.7373	0.5027	31.0005	542	True
2	2407	118.141	31.4268	3.7592	0.964	0.7314	0.0789	45.1836	688.173	True
3	704	43.5023	24.5984	1.7685	0.8248	0.8253	0.3934	22.6634	726.641	True
4	3074	97.1472	47.5979	2.041	0.8717	0.7951	0.7084	40.3224	458.502	False
5	4688	114.763	56.7615	2.0218	0.8691	0.8409	-0.5667	73.3957	882.268	False
6	477	43.1425	19.7158	2.1882	0.8895	0.6903	0.7532	160.568	566.931	False
7	996	48.8421	30.1974	1.6174	0.786	0.83	-0.3922	177.571	118.507	False
8	1527	72.7312	32.1897	2.2595	0.8967	0.7689	1.1031	215.125	119.538	False
9	791	45.4603	27.379	1.6604	0.7983	0.777	-1.1935	276.899	15.5398	True
10	885	49.967	25.4783	1.9612	0.8602	0.8157	0.1758	321.858	27.7831	False
11	865	60.9988	23.7735	2.5658	0.9209	0.67	-0.6206	306.805	864.442	False
12	9764	174.84	79.9051	2.1881	0.8895	0.756	-1.0035	341.592	370.654	False
13	5590	100.499	74.9477	1.3409	0.6662	0.8931	0.3458	390.83	706.214	False
14	772	53.4057	21.7698	2.4532	0.9131	0.7546	-1.3439	371.161	29.9093	False
15	3806	88.412	63.5037	1.3922	0.6958	0.7996	0.5855	456.034	447.32	False
16	1368	68.8013	32.6693	2.106	0.8801	0.7621	0.1327	437.326	13.0395	True
17	818	48.7035	23.1908	2.1001	0.8794	0.8503	0.5594	490.792	650.099	False
18	583	30.9882	25.7454	1.2036	0.5565	0.8807	-0.6814	486.81	163.602	False
19	6173	146.612	73.4479	1.9961	0.8655	0.639	0.4437	581.134	750.702	False
20	1268	55.1715	37.4316	1.4739	0.7346	0.792	0.9741	664.745	233.587	False
21	1408	71.6194	28.9459	2.4743	0.9147	0.8167	1.0982	723.629	987.69	True
22	3872	116.659	53.0473	2.1992	0.8906	0.6702	-1.541	793.571	878.724	False
23	3821	95.6808	52.75	1.8139	0.8343	0.9065	-1.1603	884.495	983.684	True
24	1979	71.3216	39.2045	1.8192	0.8354	0.8154	-1.307	916.707	906.786	False

25 8745 138.777 89.9937 1.5421 0.7612 0.8115 1.2047 976.542 708.753 True

Network per lacuna

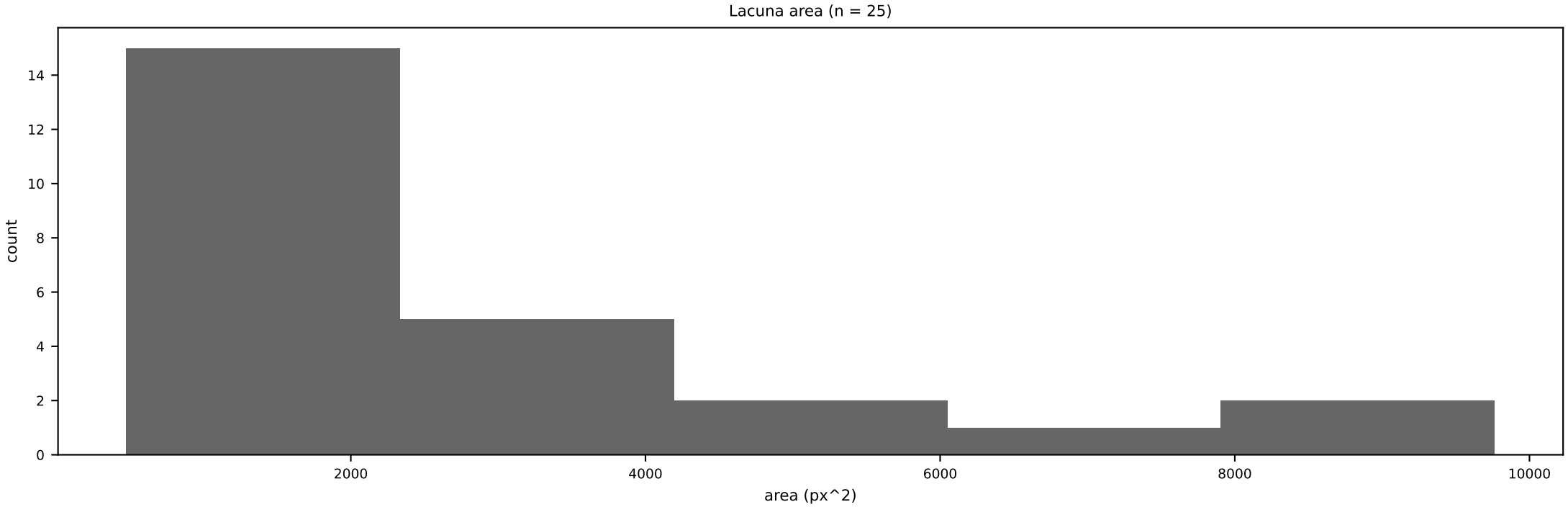
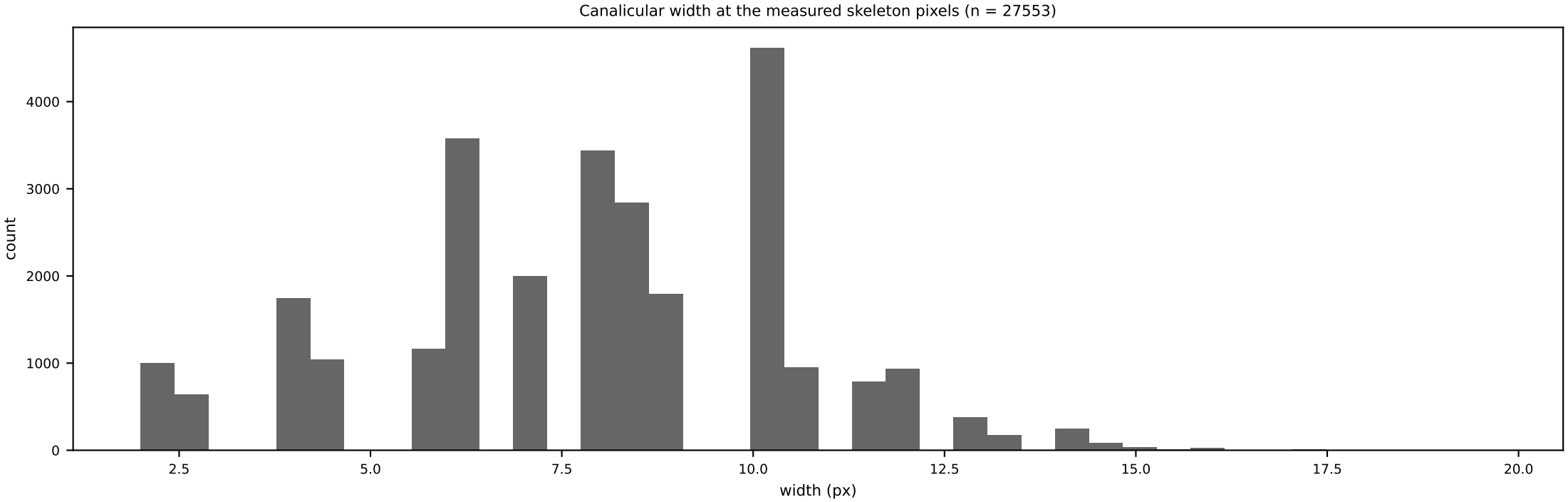
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	182	235	563.286	21	26.8232	6.9664
2	11	324	632	590.357	36	16.3988	8.0085
3	4	141	284	467.961	23	20.3461	5.7534
4	10	241	459	954.909	57	16.7528	7.7676
5	7	288	588	736.566	22	33.4803	8.7175
6	9	286	545	415.848	23	18.0803	6.9746
7	4	123	248	551.132	27	20.4123	8.0682
8	6	172	421	613.044	22	27.8656	7.5689
9	3	151	227	63.1838	2	31.5919	5.7979
10	6	70	94	75.9411	4	18.9853	6.6883
11	12	321	689	1428.11	80	17.8514	7.3687
12	24	778	1522	3355.14	183	18.3341	7.3534
13	9	469	961	2125.08	107	19.8606	8.7134
14	5	168	247	225.652	12	18.8043	7.2723
15	10	353	751	950.838	51	18.6439	7.1225
16	1	59	133	83.8406	4	20.9602	7.0119
17	4	218	399	463.345	22	21.0611	7.4299
18	7	184	566	283.451	21	13.4977	7.9669
19	22	607	1128	1944.31	98	19.8399	7.5777
20	7	274	570	813.784	44	18.4951	7.2242
21	8	311	574	1424.43	70	20.3489	7.755
22	17	525	861	1278.74	78	16.3942	7.831
23	6	201	375	404.09	23	17.5691	8.0704
24	5	175	358	319.208	19	16.8004	8.5769
25	7	428	897	763.896	40	19.0974	6.8574

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	314.877	6899	10287	0.7247	0.6734	0.0263806	0.0228444	2.2231
2	330.693	7406	15690	0.7916	0.7301	0.0437483	0.0402804	3.3263
3	132.368	3187	6317	0.7504	0.6702	0.0442422	0.044958	3.0219
4	289.563	8243	15166	0.836	0.748	0.0292369	0.0302651	3.4535
5	351.161	12196	26476	0.9903	0.8855	0.0236143	0.0222088	1.9934
6	136.853	6220	14728	1	1	0.0459807	0.0370043	6.5764
7	192.989	4006	9474	1	1	0.0307039	0.0261769	2.0727
8	221.38	6390	14742	1	1	0.0269171	0.0285579	2.7103
9	143.74	3169	5742	0.6661	0.6142	0.0476491	0.0395333	2.0871
10	148.61	2877	4028	0.8558	0.7127	0.0243309	0.0233366	4.0374
11	184.652	7467	20230	1	1	0.0429891	0.0340583	6.4987
12	750.649	18042	35989	1	1	0.0431216	0.0422907	3.1972
13	352.919	11920	25165	1	1	0.0393456	0.038188	2.5502
14	186.959	3412	6249	0.8041	0.7079	0.049238	0.0395263	2.6744
15	328.233	11363	23232	1	1	0.0310657	0.0323261	3.0466
16	193.539	4187	8710	0.6366	0.5949	0.0140912	0.0152698	0.5167
17	137.196	6367	12630	1	1	0.0342391	0.0315915	2.9155
18	104.569	5596	16799	1	1	0.0328806	0.0336925	6.6942
19	631.595	16663	33826	1	1	0.036428	0.0333471	3.4832
20	199.865	7779	21047	1	1	0.035223	0.0270822	3.5024
21	197.48	6605	13178	0.8037	0.7245	0.0470855	0.0435574	4.051
22	394.747	12775	22894	1	1	0.0410959	0.0376081	4.3066
23	281.38	6112	10015	0.7455	0.6918	0.0328861	0.0374438	2.1323

24	221.865	6537	13263	1	1	0.0267707	0.0269924	2.2536	
25	525.044	12017	25181	0.7555	0.698	0.0356162	0.0356221	1.3332	

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	173	226	197.397	261.752	4	3	2
2	259	372	377.848	734.24	5	7	10
3	122	246	165.196	327.664	4	3	3
4	188	295	275.451	520.375	5	7	4
5	278	481	336.363	685.21	9	9	10
6	243	305	329.078	630.399	3	6	7
7	123	182	138.154	271.409	4	4	2
8	152	309	194.61	474.747	4	5	6
9	70	70	173.782	259.622	1	2	4
10	70	70	78.0122	105.154	4	0	0
11	291	580	378.99	797.838	6	10	6
12	719	1338	884.767	1740.22	18	17	20
13	469	926	537.345	1087.02	9	12	8
14	160	224	189.024	283.007	5	2	3
15	257	377	394.191	861.394	11	8	17
16	39	39	61.3848	143.154	1	2	3
17	126	283	261.392	474.943	3	7	5
18	167	329	206.51	643.529	4	4	9
19	580	904	697.612	1291.34	11	16	14
20	255	480	318.907	643.872	6	8	7
21	306	518	353.25	648.63	5	8	9
22	499	821	591.002	981.436	13	17	14
23	168	295	232.551	427.333	6	6	7
24	153	207	194.782	394.978	5	4	6
25	370	622	476.747	999.394	10	12	9

548_hyp_HP_z14c1-2: distributions, pre-validation, px